

# Bowen Deng

PhD Candidate · AI for Chemistry & Materials

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## RESEARCH INTERESTS

Machine learning for chemistry and materials science: learning across molecular representations — 1D sequences (SMILES), 2D molecular graphs, and 3D geometry — with sequence models, graph neural networks, and machine-learned interatomic potentials, for reaction discovery, thermochemistry, and molecular / materials property prediction. Also interested in computer vision (human action recognition) and large language models.

## EDUCATION

### Ph.D. in Chemistry / Machine Learning

2023 – 2026 (expected)

Chimie ParisTech – Université PSL, Paris, France

Advisor: Dr. Thijs Stuyver. Thesis: AI for chemistry & materials (molecular representation learning, MLIPs, thermochemistry, reaction discovery).

### M.Sc. in Materials and Chemical Engineering

2020 – 2023

Guangxi University, Nanning, China

Advisor: Prof. Shuangliang Zhao.

### B.Eng. in Computer Science and Technology

2016 – 2020

Southwest University of Science and Technology, Mianyang, China

Double degree: second B.Eng. in Materials Science and Engineering.

## RESEARCH EXPERIENCE

### Research Intern — Computer Vision (Human Action Recognition)

2022

Institute of Automation, Chinese Academy of Sciences (CASIA)

With Dr. Dongchang Liu. Developed a fast MLP-based temporal action detection model (TadML).

## PUBLICATIONS

- [Bowen Deng](#), Thijs Stuyver. MLIP-Enhanced Thermochemistry Predictions across Organic Chemical Space. *Journal of Chemical Theory and Computation*, 22(18), 9526, 2026.
- Marco Ferrer, [Bowen Deng](#), Javier E. Alfonso-Ramos, Thijs Stuyver. Screening Diels–Alder reaction space to identify candidate reactions for self-healing polymer applications. *Digital Discovery*, 4(11), 3400–3410, 2025.
- [Bowen Deng](#), Thijs Stuyver. Graph-based deep learning models for thermodynamic property prediction: the interplay between target definition, data distribution, featurization, and model architecture. *Journal of Chemical Information and Modeling*, 65(2), 649–659, 2025.
- [Bowen Deng](#), P. Chen, P. Xie, Z. Wei, S. Zhao. Iterative machine learning method for screening high-performance catalysts for H<sub>2</sub>O<sub>2</sub> production. *Chemical Engineering Science*, 267, 118368, 2023.
- [Bowen Deng](#), S. Zhao, D. Liu. TadML: A Fast Temporal Action Detection with Mechanics-MLP. *Computer Graphics International Conference (CGI)*, 28–40, 2023.
- [Bowen Deng](#). Catalysis Distillation Neural Network for the Few-Shot Open Catalyst Challenge. *arXiv / ChemRxiv* (preprint), 2023.
- Z. Wei, [Bowen Deng](#), P. Chen, T. Zhao, S. Zhao. Palladium-based single atom catalysts for high-performance electrochemical production of hydrogen peroxide. *Chemical Engineering Journal*, 428, 131112, 2022.

## AWARDS & HONORS

### Kaggle Competitions Silver Medal x3 — Competitions Expert

2021–2022

Google Brain – Ventilator Pressure Prediction (2021);

Hungry Geese (2021);

Feedback Prize – Evaluating Student Writing (2022)

## ACADEMIC SERVICE

Reviewer, *Digital Discovery*

Reviewer, ACDSA 2027 (IEEE)

## TECHNICAL SKILLS

|                        |                                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>     | Python, C/C++, CUDA, Triton                                                                                   |
| <b>ML / DL</b>         | PyTorch, PyTorch Geometric, e3nn, graph neural networks, machine-learned interatomic potentials, transformers |
| <b>Cheminformatics</b> | RDKit, SMILES, high-throughput reaction profiling, quasi-RRHO thermochemistry                                 |
| <b>Comp. chemistry</b> | Gaussian, ORCA, VASP, CP2K, LAMMPS, xTB (DFT / ab initio / MD)                                                |
| <b>Tooling</b>         | Git, Linux, SLURM / HPC clusters                                                                              |
| <b>Languages</b>       | Chinese (native), English (fluent)                                                                            |